



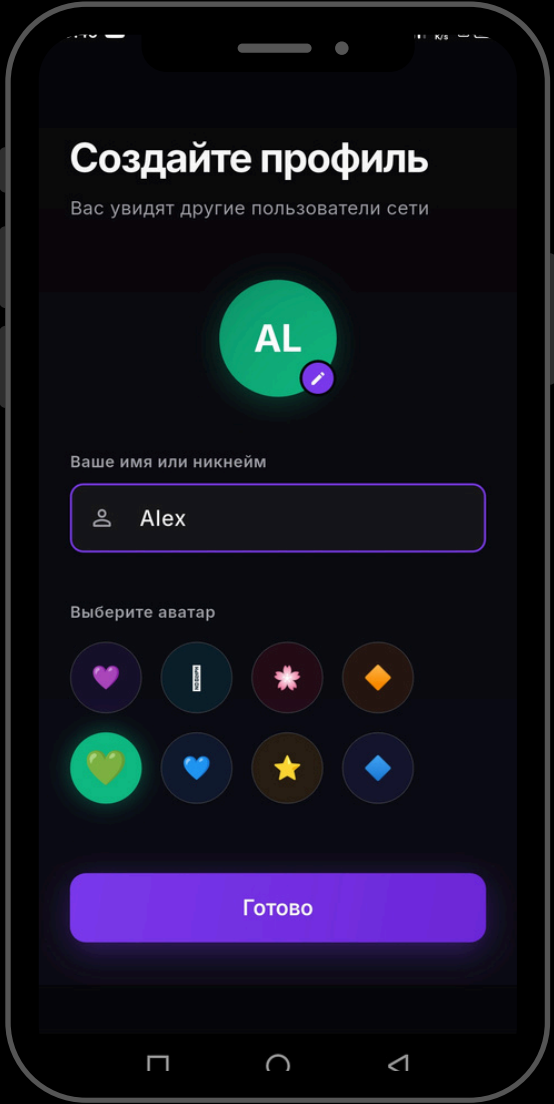
СИСТЕМА
ДЕЦЕНТРАЛИЗОВАННОЙ
СВЯЗИ



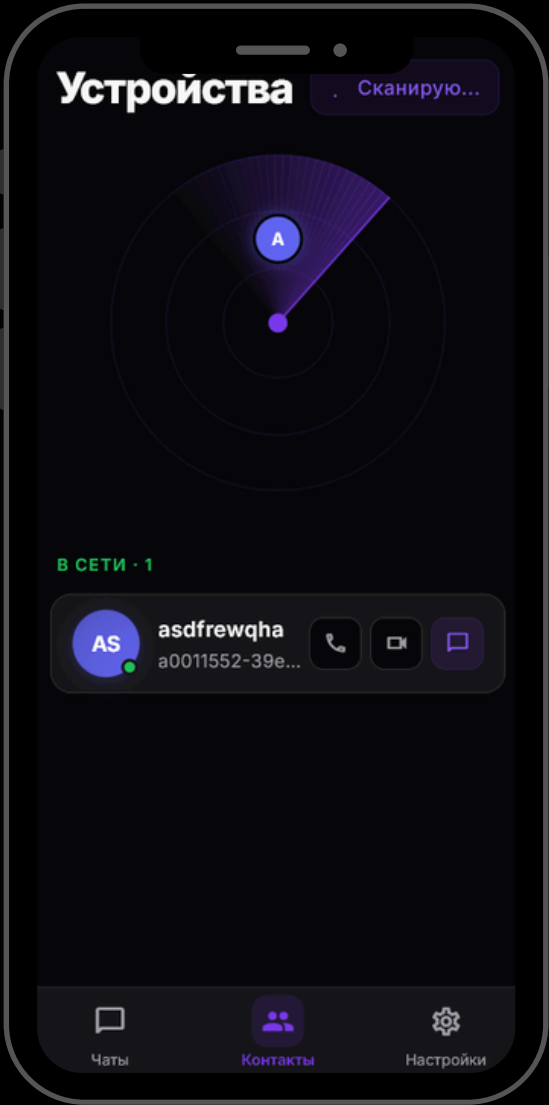
Кроссплатформенность



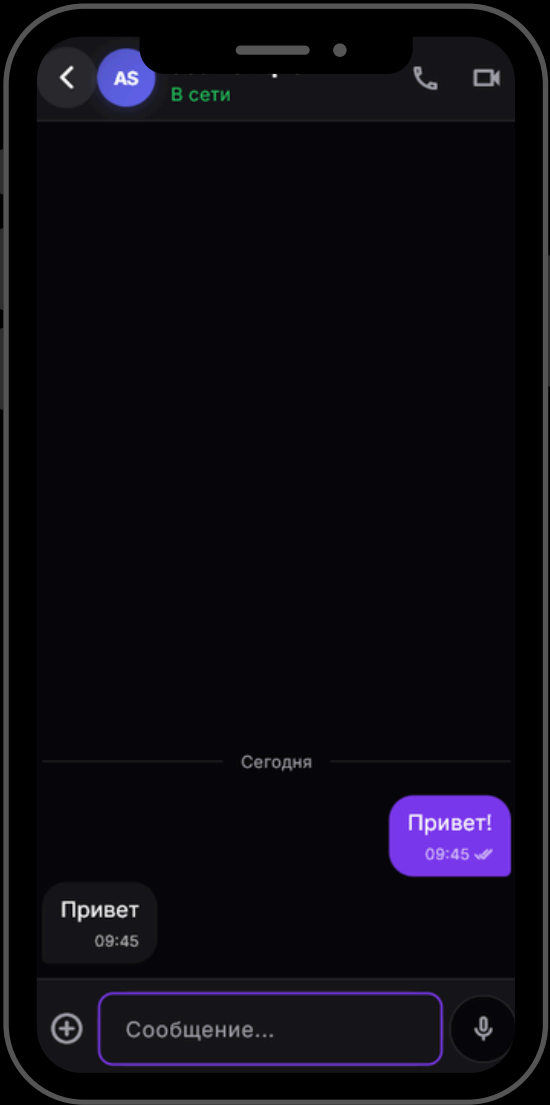
EXPERIMENTAL



Создание
пользователя



Поиск пиров



Двусторонний
обмен



ВОЗМОЖНОСТИ ПОЛЬЗОВАТЕЛЯ

[ТЗ]

Текстовые сообщения



Групповые чаты



Отправка файлов



Просмотр медиа



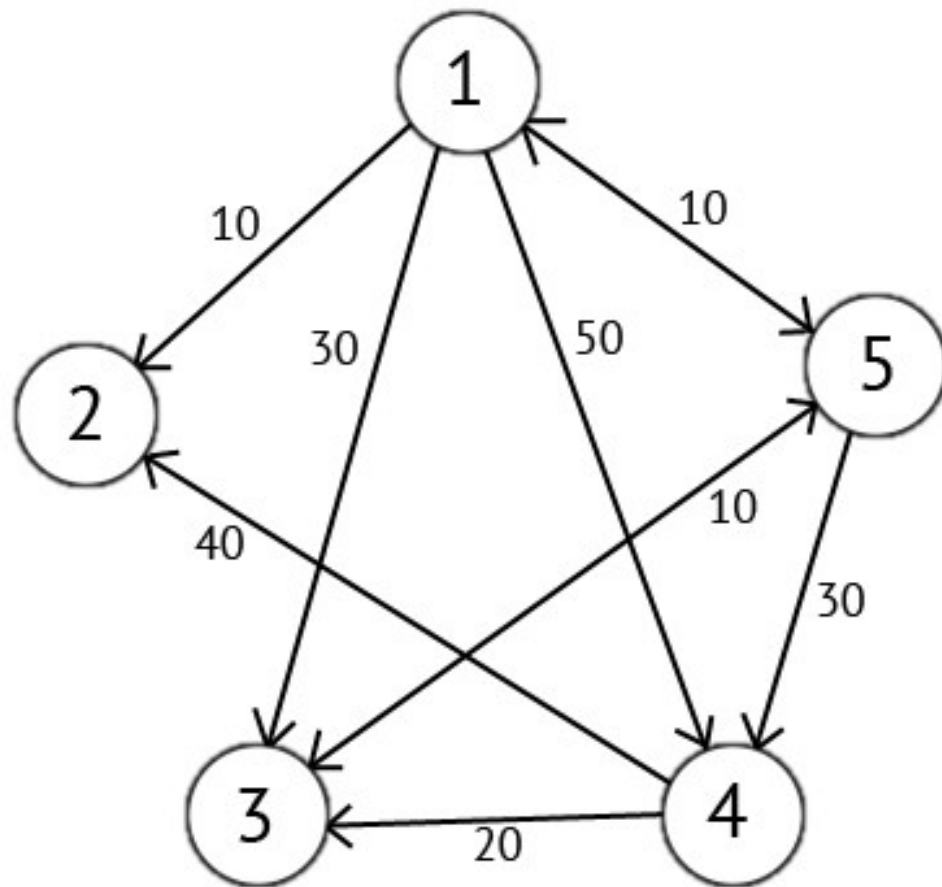
Звонки



Персональная настройка



Архитектура



mDNS/DNS-SD
UDP broadcast



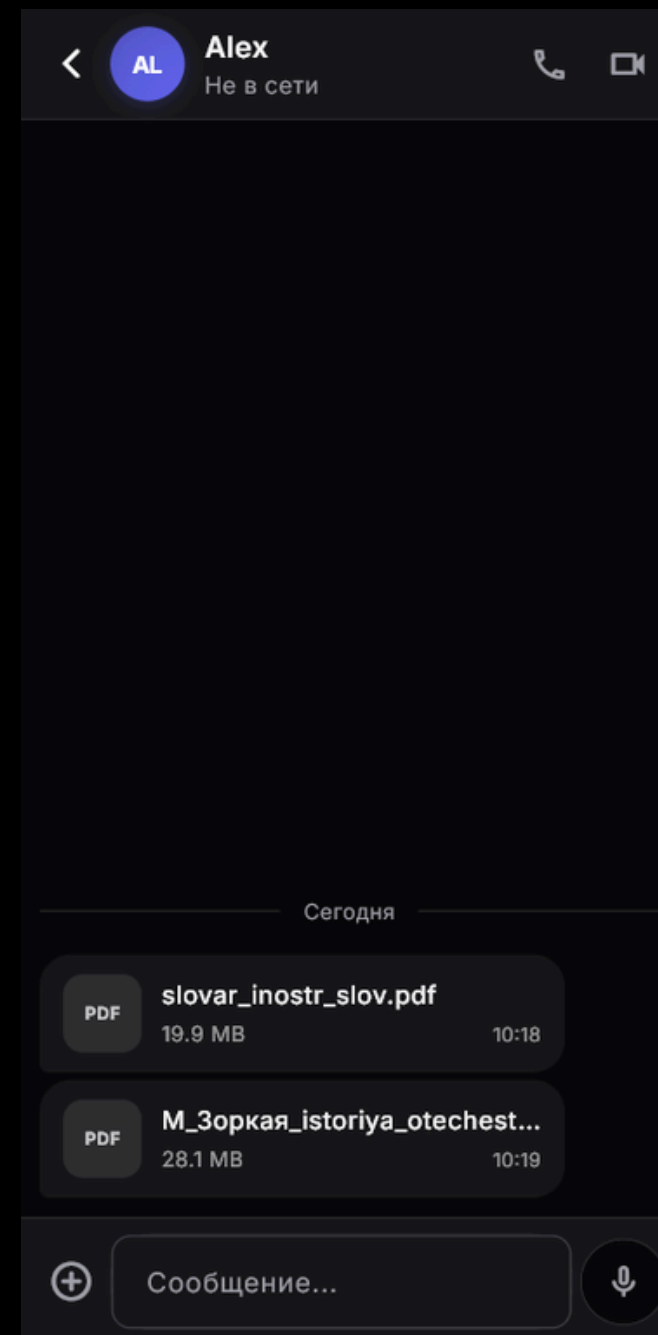
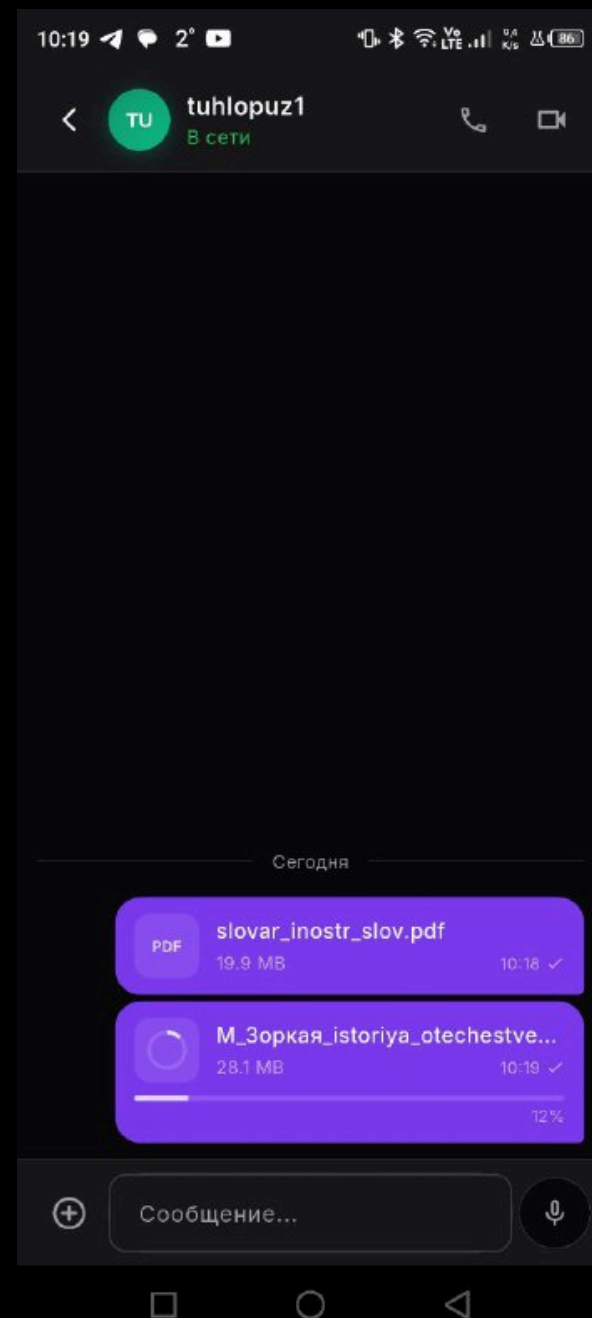
Защита от петель: Mesh-пакет
содержит hop-list
Ограничение количества хопов до 5



Смешанная классическая p2p
архитектура и mesh сеть
(WIFI Direct)



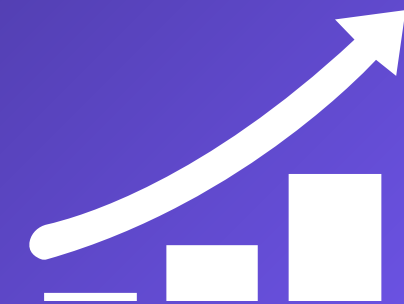
Передача файлов



Чанки по 512kb, зашифрованные
ключом p2p (ChaCha20)

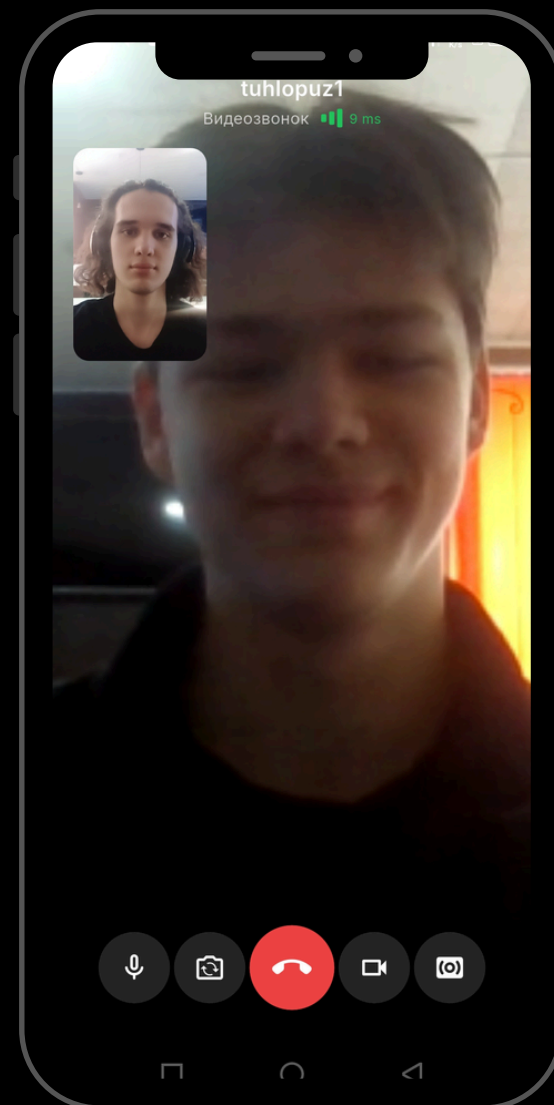
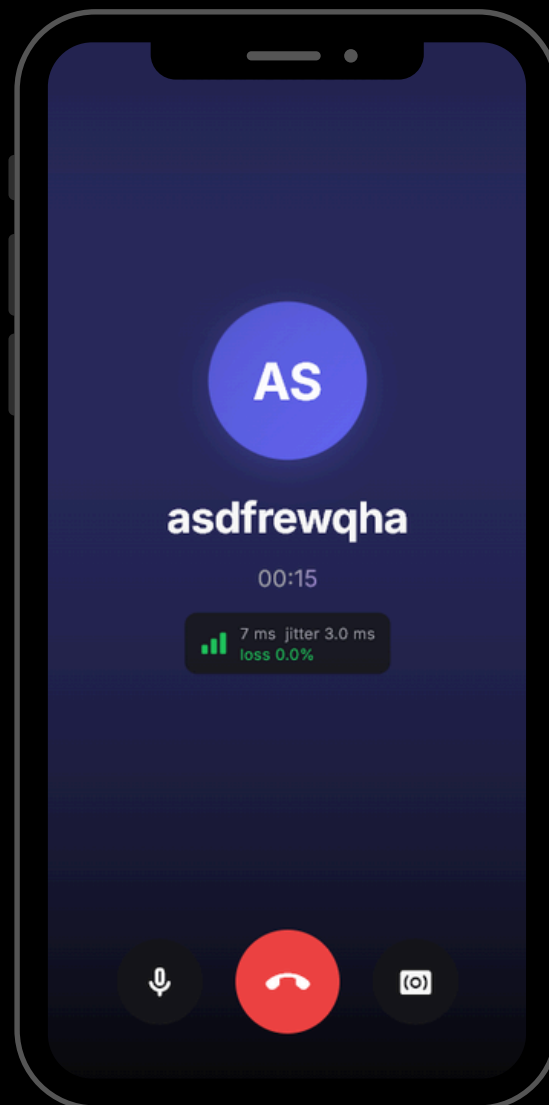
SHA-256 - хеш полного файла
(контроль целостности)

ограничение
нагрузки **5mb/s**



Real-time звонки

реализованы с помощью
WebRTC





Метрики и стабильность ЗВОНКОВ



Во время звонков приложение
отслеживает качество соединения и
выводит информацию в реальном
времени.

при плохой связи - в видеозвонке
отключается видео



0 ms jitter 2.0 ms
loss 0.0%



Безопасность

Защита от спуфинга (zero trust)

UUID + pubkey binding при первом подключении, hello-handshake перед любым обменом данными

Аутентификация

X25519 ECDH

шифрование трафика

ChaCha20-Poly1305 AEAD

меры предприняты против спама.

Ограничение 60 сообщений/10 секунд на пользователя



Надёжность

Keep-alive
каждые 30с -
для
поддержани
я связи

Dedup по id,
но CSPRNG
nonce на
каждое
сообщение

Рероутинг
при потере
связи

TSP -
гарантия
порядка
сообщений
АСК для
файлов



Tech stack

- Go 1.22 — сетевой движок. Горутины, бинарник ~8 MB, GC latency <1 ms, быстрее аналогов
- Flutter — один код для Windows, Android, iOS, Linux, macOS (кроссплатформенность), простота разработки
- CGo FFI — Go и Flutter говорят напрямую, без HTTP и JSON, нет необходимости TCP прослойки между беком и UI
- X25519 + ChaCha20-Poly1305 — E2E-шифрование из коробки
- Custom P2P mesh — Dijkstra + gossip + CRDT, без центральных серверов, лучше оптимизация под LAN
- mDNS + BLE — обнаружение соседей без долгой настройки инфраструктуры

Backend



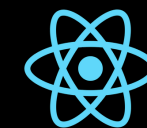
Frontend



P2P



Web Site



Storage

sqlite



Наша команда



НИКИТИН ВЛАДИМИР



ВОРОБЬЕВ ФЁДОР



УДАЛОВ МАКАР



БАРМИН СТЕПАН

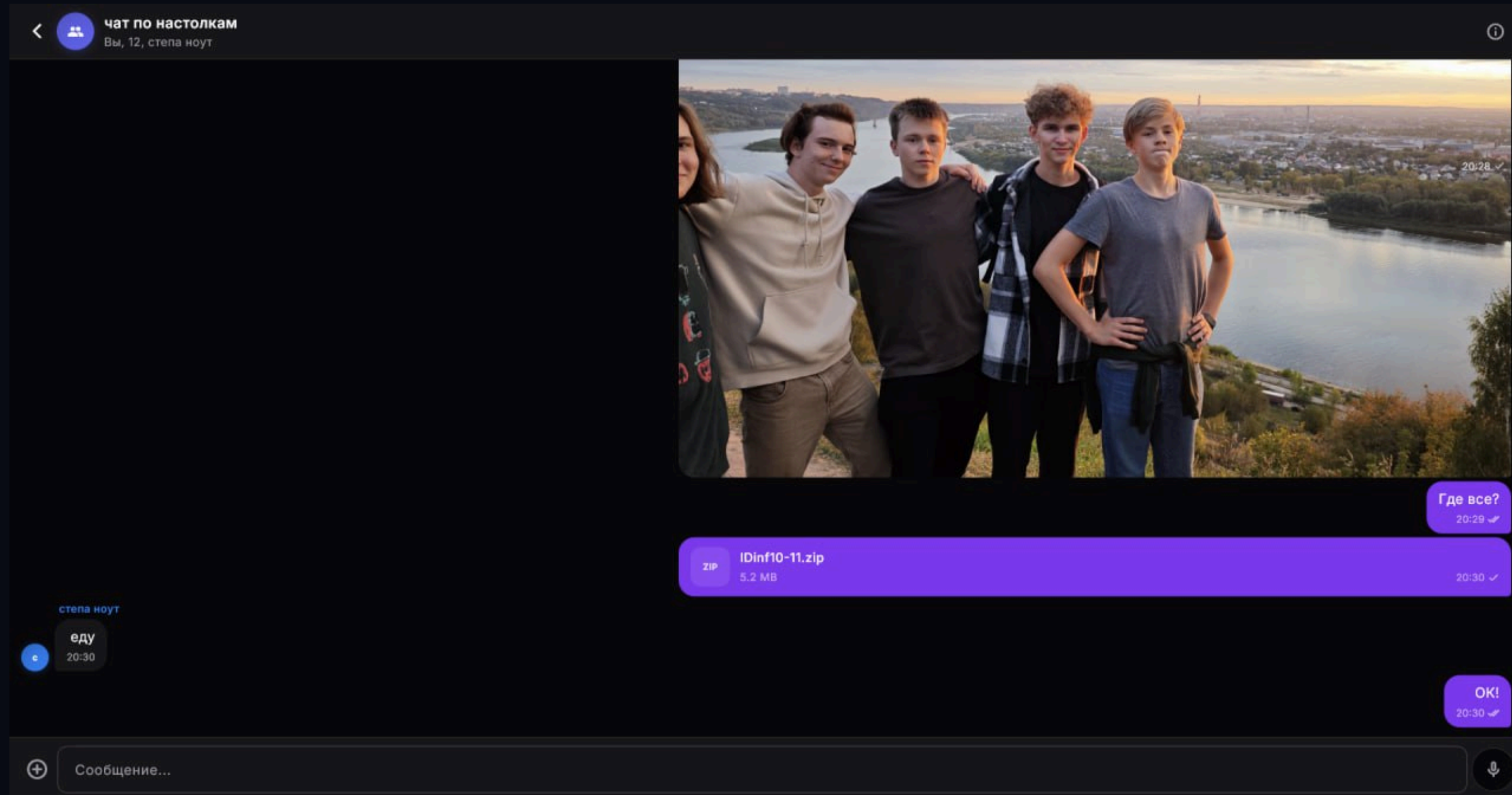


Больше скриншотов



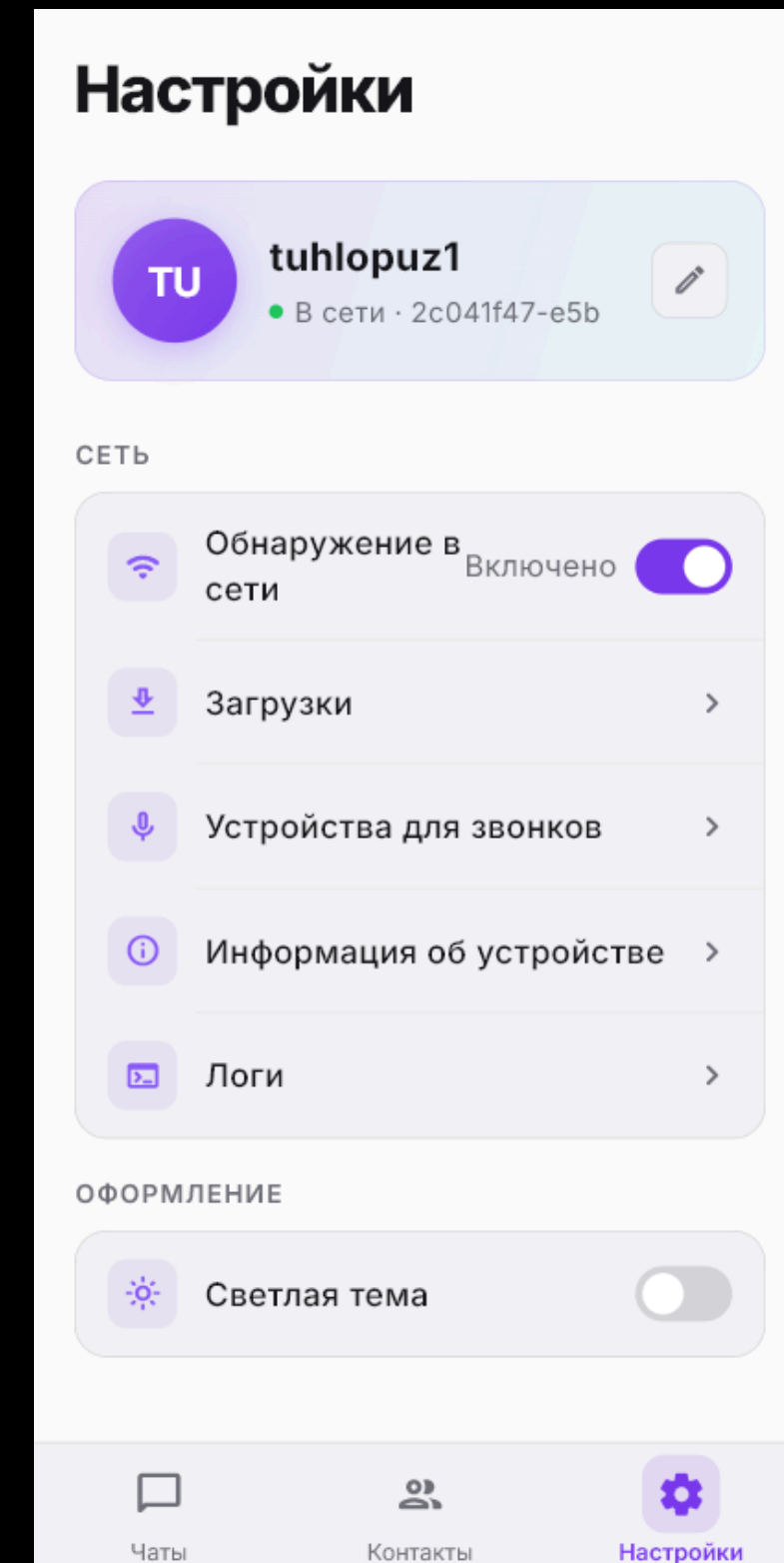
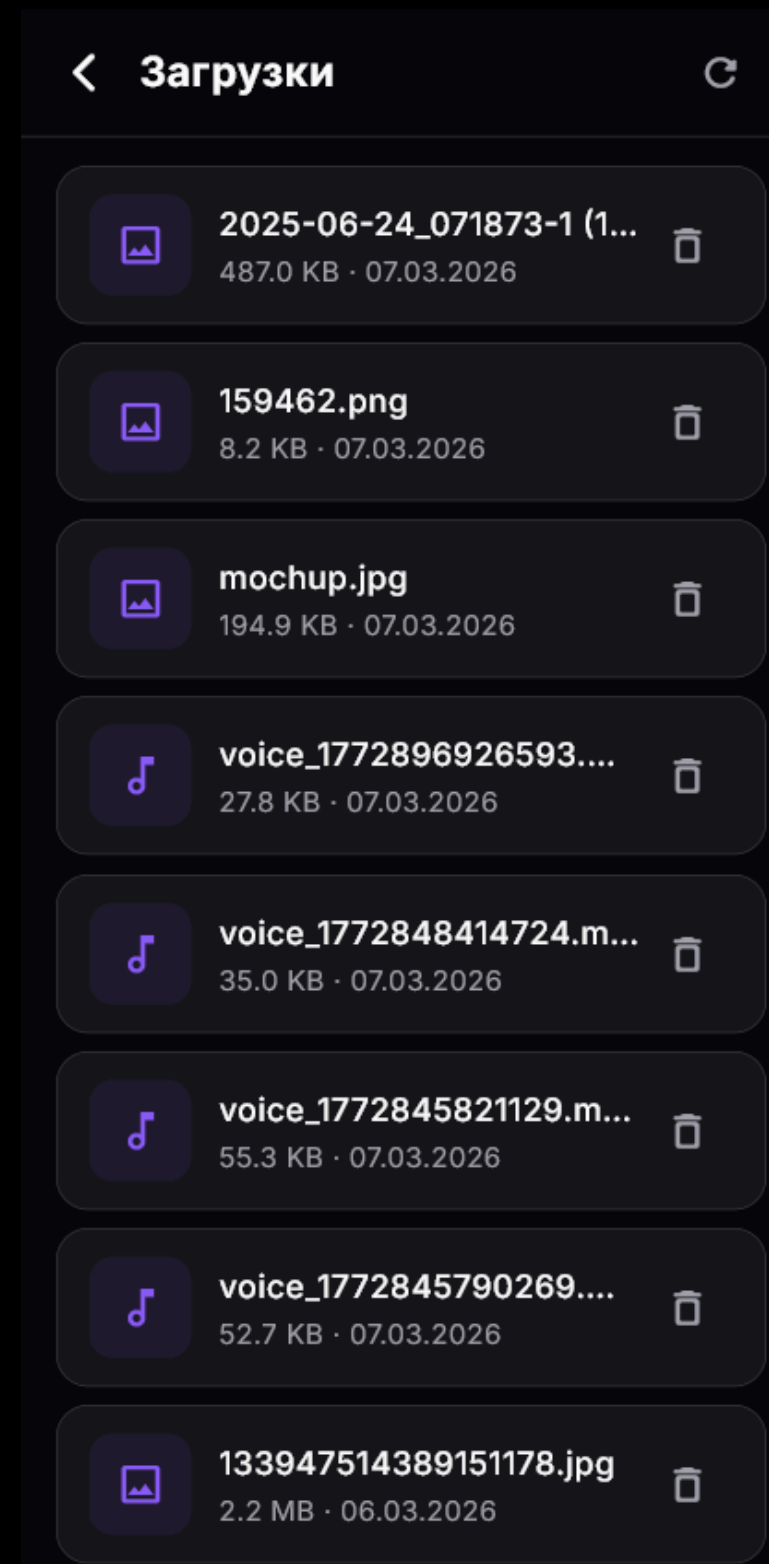
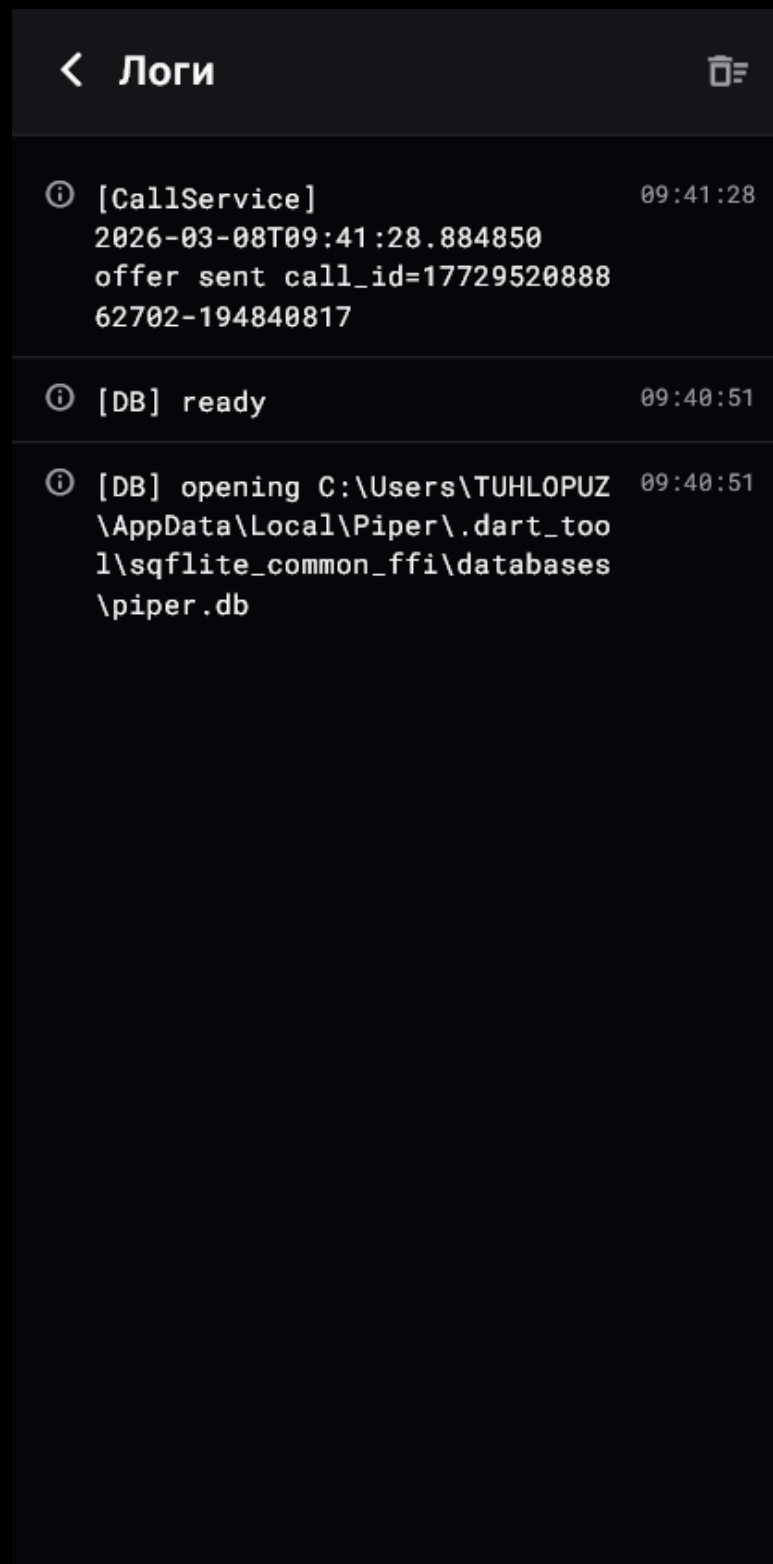


Больше скриншотов





Больше скриншотов





Больше скриншотов

Summary

All jobs

Go Tests

Linux Release

macOS Release

Windows Release

Android Release

Publish GitHub Release

Run details

Usage

Workflow file

Go Tests

succeeded 13 hours ago in 31s

Run tests

202 --- PASS: TestPeerManager_ConcurrentAccess (0.00s)

203 === RUN TestSpamBucket_AllowsUnderLimit

204 --- PASS: TestSpamBucket_AllowsUnderLimit (0.00s)

205 === RUN TestSpamBucket_DropsOverLimit

206 --- PASS: TestSpamBucket_DropsOverLimit (0.00s)

207 === RUN TestSpamBucket_ResetsAfterWindow

208 --- PASS: TestSpamBucket_ResetsAfterWindow (0.00s)

209 === RUN TestNode_AllowFromPeer_IndependentBuckets

210 --- PASS: TestNode_AllowFromPeer_IndependentBuckets (0.00s)

211 === RUN TestTransferManager_Start

212 --- PASS: TestTransferManager_Start (0.00s)

213 === RUN TestTransferManager_Start_Receiving

214 --- PASS: TestTransferManager_Start_Receiving (0.00s)

215 === RUN TestTransferManager_Get

216 --- PASS: TestTransferManager_Get (0.00s)

217 === RUN TestTransferManager_UpdateProgress

218 --- PASS: TestTransferManager_UpdateProgress (0.00s)

219 === RUN TestTransferManager_Complete

220 --- PASS: TestTransferManager_Complete (0.00s)

221 === RUN TestTransferManager_Fail

222 --- PASS: TestTransferManager_Fail (0.00s)

223 === RUN TestTransferManager_Remove

224 --- PASS: TestTransferManager_Remove (0.00s)

225 === RUN TestTransferManager_List

226 --- PASS: TestTransferManager_List (0.00s)

227 === RUN TestTransferManager_ConcurrentAccess

228 --- PASS: TestTransferManager_ConcurrentAccess (0.00s)

229 PASS

230 ok github.com/catsi/piper/core 3.642s

> Post Set up Go

> Post Checkout

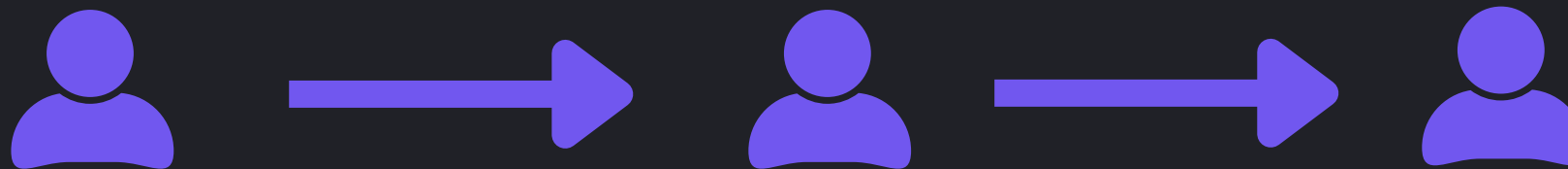
> Complete job



**Спасибо
за
ВНИМАНИЕ**



Мультихоп



Алгоритм маршрутизации

Source routing: Dijkstra
по weighted graph
($RTT + loss^2 \times 10000 +$
тип канала)

дедупликация

защита от петель

MESH СЕТЬ



< **Устройства для звонков**

 МИКРОФОН

Microphone (High Definition Audio Device) ✓

 ДИНАМИК

Динамики (High Definition Audio Device) ✓

 КАМЕРА

HD Webcam ✓

Как это работает

Finance Division Overview >

Financial
Planning
and
Analysis



Accounting
and
Compliance



Cost
Manage-
ment



Treasury
and Risk
Mana-
gement



Perfor-
mance
Measure-
ment





Killer feature #1

MESH CETЬ



vision

To be a global leader in the textile industry, recognized for our commitment to quality, innovation, and sustainability, while enhancing the lives of our customers and communities.



mission

At Ginyard International Co., our mission is to provide superior textile products that meet the highest standards of quality and sustainability. We aim to foster strong partnerships with our clients, invest in innovative technologies, and uphold ethical practices in all aspects of our operations, ensuring long-term growth and value for our stakeholders.

highest standards
of quality







Killer feature #2

АНАЛИТИКА

Investment opportunities are critical to fostering growth and innovation within the company. Our analysis focuses on areas that align with our strategic goals.

1. Capital Expenditures (CapEx)
Upgrading machinery to enhance production efficiency and reduce waste.

2. Research and Development (R&D)
Focused on developing eco-friendly fabrics and innovative textile solutions.

3. Market Expansion
Targeted campaigns to enter new geographic markets.



Killer feature #2

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Policy implication

committed to delivering high-quality fabrics and innovative textile solutions. Established in 1986, we have built a reputation for excellence in the industry, leveraging advanced technology and sustainable practices to meet the diverse needs of our clients. Our extensive product range includes cotton, polyester, blended fabrics, etc. Serving various sectors such as fashion, home furnishings, and industrial applications.



conclusion

In summary, the Finance Division of Ginyard International Co. has demonstrated strong financial performance and strategic foresight throughout the fiscal year 2024. Our comprehensive financial forecasting indicates a positive trajectory for revenue growth and profitability, driven by effective cost management and targeted investments in innovation and market expansion.

We are committed to navigating the challenges of economic volatility, competition, and regulatory compliance through proactive strategies that ensure our financial stability and long-term success. By continuing to optimize our cash flow management and investing in sustainable practices, we position ourselves for resilient growth in an ever-evolving market.

As we move forward, our focus will remain on aligning our financial objectives with the overall strategic vision of the company. By fostering a culture of accountability and continuous improvement within the Finance Division, we aim to support Ginyard International Co. in achieving its goals while delivering value to our stakeholders.

